Aligned equations

Several equations can be aligned on their relation signs:

$$(a+b)^2 = a^2 + 2ab + b^2$$
$$(a-b)^2 = a^2 - 2ab + b^2$$

and a definition by cases uses a brace on the left:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{otherwise} \end{cases}$$

A matrix closes the fixture:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$